CHAPTER 1: INTRODUCTION

The dim luster of a chest X-ray has been long a physician lantern in the dark, showing dim shadows of disease in the lungs. However, in most parts of the world, that lantern glows low-wattage, constrained by the number of trained radiologists, human weariness in interpreting and the delicacy with which diseases like pneumonia and tuberculosis tend to manifest themselves. These respiratory diseases continue to be one of the long-standing health burdens around the world leading to immense death and suffering, especially in those areas where healthcare facilities are on edge. Delayed or misdiagnosis according to the recent studies remains a hindrance to timely treatment and containment of infectious diseases, which further increases their effects on the society (Elhanashi et al., 2025). Although the cost-effective and widely available, chest radiography requires a certain level of expertise and consistency that is not always present and even the most experienced clinicians can disagree on their interpretations in case of the ambiguous visual pattern.

As a landscape, the intersection of artificial intelligence and medical imaging does not appear as a technological development, but as a development that must occur. Deep learning, especially convolutional neural networks, have been shown to be capable of extracting complex patterns in visual data and is frequently fast and more accurate than the classical rule-based methods. Modern neural architectures are trained on data, unlike previous computational methods which heavily depended on handcrafted features which are sensitive to subtle changes in texture, intensity, and structure that are imperceptible to the human eye (Rayan et al., 2025). This is particularly useful in the analysis of chest X-rays, where disease manifestations are early, and may be barely noticeable. Nonetheless, most deep learning models are opaque systems and provide predictions without understanding how they reach such predictions; this has created an issue of trust and uptake by the medical community (Mohammed, 2024).

The current work addresses these issues by presenting an AI-based chest X-ray disease detection system that does not only focus on the accuracy, but also strives to be as interpretable as possible. At the very heart of this is EfficientNet architecture, a well-balanced convolutional neural network that harmoniously scales the depth, width and resolution. This design can allow the model to be highly performing and yet be computationally efficient thus it is applicable in the resource-constrained environment. The capabilities of EfficientNet in order to extract both global and local features guarantees the recognition of subtle pathological features (e.g. lung opacities, consolidations, or abnormal textures) when performing classification tasks (Das et al., 2025). Complementing this architecture is the integration of Gradient-weighted Class Activation Mapping, a technique that transforms abstract predictions into visual explanations. Grad-CAM helps fill the divide between machine inference and clinical judgment by pointing to the region(s) of an X-ray that a model has considered to make its decision, thus enabling a practitioner to assess whether the model is focusing on clinically-relevant regions or not.

Such system is not only necessary due to curiosity in technology, but it is needed practically. The patient to radiologist ratio is more than normal in most healthcare facilities, particularly the rural clinics and under-served areas. The imbalance frequently leads to the delayed diagnosis, overworking of medical personnel, and a greater risk of making a diagnosis mistake. This can be relieved by automated systems, which offer quick initial examinations so that clinicians can give more focus to urgent cases and make effective decisions by prioritizing the urgent cases. Additionally, explainable AI guarantees that such systems are not operated as black boxes but as a means of collaboration among humans and as a way to augment human knowledge. According to modern literature, AI-based diagnostics are much more likely to gain the trust of clinicians when it is transparent and makes the implementation of AI in clinical practice easier (Nwaiwu and Das, 2024).

The main aim of this project is to create a strong and interpretable deep learning model that will be able to classify chest X-ray images into three clinically significant categories namely normal, pneumonia as well as tuberculosis. This is achieved by a systematic pipeline that involves a preprocessing of data, data augmentation, training of the model and evaluation, after which visual explanations are generated using Grad-CAM. Through transfer learning with EfficientNet, the system takes advantage of the knowledge that is already trained but it is able to fit the particularities of medical imaging. The model will not only be highly accurate, but also consistent across different datasets, thus ensuring consistency in its performance in real-life applications.

CHAPTER 2: PROFILE OF THE PROBLEM

2.1 Problem Statement

In the broad avenues of contemporary healthcare, the process of deriving meaning out of the images of the chest X-rays is indispensable and very human.To these long-term challenges, the need to have automated diagnostic systems comes out in a very irresistible manner.

Although the idea of artificial intelligence into medical imaging has potential, there are some challenges. Availability and quality of data are one of the very urgent. Medical records can be small, unbalanced, or inconsistent and the imaging conditions, equipment and patient demographics vary across records. This variability may impact on model performance, which will result in poor generalization, once applied to the real world. Moreover, annotating medical images involves human effort and, hence, this procedure is time-consuming and resource-intensive when it comes to creating a large-scale dataset (Das et al., 2025).The other important issue is interpretability of AI models.

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2.2 Scope of Solution.

The proposed solution will be able to fill the gap between the technological capability and clinical necessity in this complex environment. It will be an automated, precise and interpretable system to detect the lung diseases based on the image of the chest X-ray. The model is able to balance performance and computational efficiency by utilizing the EfficientNet architecture, which is why it can be deployed in a high-resource environment, as well as in limited-resource environments, such as advanced hospitals. The system can be further improved with the integration of Grad-CAM which will provide the visual explanations of the model predictions, thus, overcoming the issue of transparency and trust.

This solution can be classified further than that. It includes the whole diagnostic pipeline, including image preprocessing and augmentation, model training, evaluation and deployment. The system can differentiate between normal, pneumonia, and tuberculosis cases and give clinicians actionable insights that can be used to help in the early intervention. Its compact nature can be implemented on handheld devices, and thus can be used in telemedicine platforms, and in rural healthcare facilities where existing diagnostic facilities might be scarce.

Moreover, the solution is scalable, and one can expand it in the future and include other diseases and imaging modalities. The system can be optimized and its clinical applicability increased with the increasing datasets and models, making it more accurate and effective. This way, it is not a definite endpoint but a starting point where more all-encompassing diagnostic instruments can be created.

Table 2.1: Key Challenges in Traditional and AI-Based Chest X-ray Diagnosis

Aspect	**Traditional Diagnosis**	**AI-Based Diagnosis**
Accuracy	Depends on radiologist expertise	High with trained models
Speed	Time-consuming	Rapid prediction
Consistency	Variable across clinicians	Consistent outputs
Availability	Limited in rural areas	Scalable and accessible
Interpretability	Human reasoning	Requires explainability tools

CHAPTER 3: EXISTING SYSTEM

3.1 Overview of Existing AI-Based Diagnosis Systems

During decades the art of medical image interpretation has been based on the trained eye of the radiologist- an art that has been developed over time, by experience, intuition and clinical knowledge. As the era of artificial intelligence has arrived, though, this conventional method has started to evolve to a cooperative interaction between human knowledge and computational accuracy. Current AI-driven diagnosis systems within the healthcare sector and specifically in the field of the chest X-ray analysis, have been created to help clinicians in diagnosing diseases like pneumonia, tuberculosis, and other lung abnormalities. These systems are designed to save on diagnostic time, enhance accuracy, and meet the increasing need of medical imaging interpretation both in the developed and resource-limited settings.

Initial AI systems used classical machine learning methods, in which manually-derived image features, like texture descriptors, edge detector, histogram features, etc.

3.2 CNN Models in the healthcare.

Most modern medical imaging systems are based on the architecture of the Convolutional Neural Networks. The images based tasks are uniquely performed with such models because they can capture the spatial hierarchies with the help of convolutional layers, pooling, and non-linear activations. With time, various CNN architectures have been modified and optimized to healthcare tasks, all of which have helped to improve automated diagnosis.

One of the earliest models that was influential is the VGGNet that had a deep and homogenous architecture (consisting of stacked convolutional layers). Despite being computationally expensive, VGG showed that network depth could be increased to a great extent without negatively affecting image classification accuracy. Expanding on this concept, ResNet proposed a type of residual connections, so networks can be made much deeper without experiencing a vanishing gradient issue.

3.3 Explainable AI as a part of Current Systems and limitations of existing systems.

Although CNN based systems have been very accurate, the use of CNN based systems has been slowed in clinical practice due to concerns about interpretability.

Although there are considerable improvements, the current AI-based diagnosis systems also have their flaws. The reliance on big and high-quality datasets is one of the main restrictions. Medical imaging datasets are usually rather small, and can be under-represented in terms of classes, with some disease types becoming underrepresented. This may give rise to biased models, which are good working on majority classes but bad in minority conditions, thus decreasing the overall reliability (Elhanashi et al., 2025).

The other difficulty is the problem of generalization.

Large deep learning models, especially large architecture models, demand both large processing power and memory. Even though such architectures as EfficientNet have been able to deal with this problem to a certain degree, deploying AI systems in low-resource environments is still problematic.

The morality of the situation makes it even more complicated. Data privacy, bias in the algorithms, and accountability are the issues that should be properly considered. The application of patient data to train AI models must be under stringent regulations and any bias in the data may result in the uneven performance of AI models in various groups of populations.

Table 3.1: Comparative Analysis of Existing CNN Models in Healthcare

Model	**Key Feature**	**Advantage**	**Limitation**
VGGNet	Deep architecture	High accuracy	High computational cost
ResNet	Residual connections	Enables deeper networks	Complex structure
DenseNet	Dense connectivity	Efficient feature reuse	Memory intensive
EfficientNet	Compound scaling	High accuracy with fewer parameters	Requires fine-tuning
CheXNet	DenseNet-based model	Radiologist-level performance	Limited interpretability

The history of the development of AI-based diagnosis systems can be seen as a gradual shift to intelligent automation compared to manual interpretation. Transfer learning-based CNN architectures with the help of explainable AI techniques have greatly increased the accuracy and efficiency of medical image analysis. However, the trip is not over yet. Lack of data quality, generalization, interpretability and ethical issues have remained a limiting factor as to what these systems can accomplish.

This current landscape thus offers a base and a challenge. It provides well-tested methodologies and architectures that show how AI can be used in healthcare, but also point out the open areas that need to be filled to realize trustworthy and scalable solutions. The suggested system is based on this premise as it aims at incorporating effectiveness, precision, and readability into a unified system that would be in harmony with the real-world requirements of the modern healthcare setting.

CHAPTER 4: PROBLEM ANALYSIS

The idea of a smart diagnostic system is not just a technical game; it is a reaction to an lived situation in which time and expertise as well as access are unequally distributed. The imbalance is particularly apparent in the sphere of chest X-ray interpretation, where clinicians might be asked to work with large quantities of images on a time-sensitive basis, and the outcomes of such work are directly related to patient outcomes. The current project is the result of this tension as it aims to translate the developments of artificial intelligence into a system, which is not only accurate but also practical, available, and in line with clinical workflows. The following analysis is an introspection into the nature of the product, the possibility of its actualization and the pre-planned process by which the product is actualized.

In its simplest form, the proposed product is an AI-based diagnostic support system that is aimed at classifying the images of chest X-rays into clinically meaningful groups, i.e. normal, pneumonia and tuberculosis. The system is not constrained by deterministic logic as in the traditional software systems, but rather the learned representation that is based on data shapes the system. It is a model that is based on the EfficientNet architecture, and it is trained to identify the patterns in the radiographic images, which are then converted into probabilistic predictions. The interpretability is an intentional focus of the product, which makes it stand out of a plethora of existing solutions. The system is not only able to provide a classification result through the integration of Grad-CAM but it also shows the parts of the image that contributed to its decision. This two-fold output, prediction with visual clarification, places the system as a collaborative one, or one that enhances clinical reasoning and not clouds it.

Product is also characterized by the flexibility and deployment aspects. It is also meant to be used in different computational settings, such as high-performance hospital systems as well as lightweight devices in remote healthcare. The system recognizes the limitations of the real world implementation by optimizing it to be more efficient and allowing it to be deployed using frameworks like TensorFlow Lite. The interface, which is envisioned by a simple web based platform enables the clinician or healthcare workers to post an X-ray image, get a diagnosis, and view the heatmap of the image without possessing any special technical expertise. In this regard, the product does not exist in isolation in the laboratory, but it is formed in a way that it is supposed to be used practically in the field and in this context, clarity, speed and reliability are important.

It is based on the three dimensions of feasibility of such a system: technical, operational and economic. Technically, the project is based on the well-known deep learning schemes and utilizes transfer learning to overcome the drawback of the size of the datasets. The fact that pre-trained EfficientNet models do not require a lot of computational resources to train, but still enable the system to be highly accurate, indicates the effectiveness of the methodology. The libraries of the modern world, like TensorFlow and Keras, offer powerful algorithms to develop, train and evaluate models, thus making sure that the technical background of the project is both stable and scalable. Moreover, the use of the data augmentation methods helps to overcome the issue of overfitting and increase the generalizability of the model in a variety of imaging conditions. Technical feasibility is thus not just due to the maturity of the available tools but also due to the prudent design decisions that are in line with the limitations of medical data.

Operational feasibility takes into account the integration of the system into the real-life healthcare settings. Even a highly advanced diagnostic tool should be within the bounds of routines and expectations of users. In this aspect, the proposed system will be user friendly and simple to use. The process of uploading the image, generating the prediction and visualizing the heatmap is based on the simple steps that the clinicians are used to when learning the radiograph and interpretations. The system lowers the barrier to adoption by ensuring they are as simple to use as possible. What is more, it can be easily integrated because it can be used as a decision-support tool, but not instead of human judgment, which guarantees that it does not interfere with the current clinical practices. The fact that explainable outputs are included also contributes to operational feasibility since it enables clinicians to confirm the reasoning of the system and the ability to stay confident in using it. The system could be used as a mentoring hand in the workplace where the experience is scarce, and the preliminary evaluation is consistent and can be used to make timely decisions.

Economic viability, which is not an issue of concern in the technical debate, is a decisive factor as to whether a system can go beyond conceptual demonstration. The solution offered is such that the costs incurred in the development, deployment and maintenance of the solution are minimized. The project can be deployed without proprietary software or large-scale computing infrastructure due to the use of the open-source frameworks and models that have been pre-trained. The EfficientNet architecture is lightweight and optimizes the models, which allows it to be deployed to devices with a limited processing power, saving the cost of purchasing costly hardware. Such considerations are of core concern in healthcare systems with limited resources. The fact that it can provide a functional diagnostic tool without the need to put a lot of financial strain implies that chances of its real-life application are high.

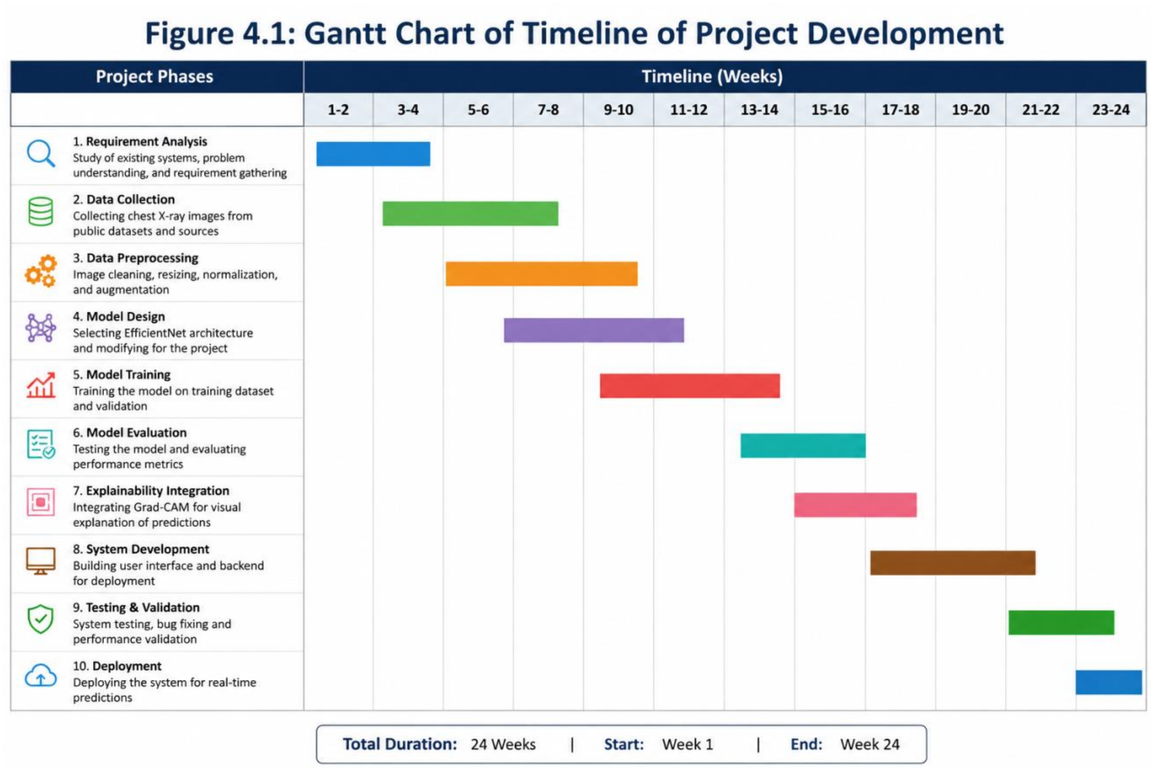


Figure 4.1: Gantt Chart of timeline of Project Development.

This plan is coherent in that it is iterative in nature. The stages influence the other stage, as well as provide feedback and refinement. This solution takes into consideration that the creation of an AI-based system is not a linear process, but rather adaptive and needs to be evaluated and changed on a regular basis.

Table 4.1: Feasibility Analysis of the Proposed AI-Based Diagnostic System

Feasibility Type	**Key Considerations**	**Outcome**
Technical	Availability of deep learning frameworks, EfficientNet architecture, dataset preprocessing methods	Highly feasible due to mature tools and proven models
Operational	Ease of use, integration with clinical workflow, explainability through Grad-CAM	Feasible with minimal training required for users
Economic	Use of open-source tools, low hardware requirements, scalable deployment	Cost-effective and suitable for resource-limited settings

When considering this analysis, it is clear that the power of the proposed system will be not only in its technical complexity but also in its correlation with practical demands. It is informed by a consciousness that innovation to be significant should be anchored on practicality. Its conception is clear, supported by a well-thought design, and the development process is based on a detailed plan.

CHAPTER 5: SOFTWARE REQUIREMENT ANALYSIS

5.1 System Overview

The suggested system is an unobtrusive yet meaningful co-worker of the clinician, which is not aimed to substitute the human judgment but to enhance and assist it. The software is about AI-based diagnosticsge model that takes an image of the chest X- ray and gives a result and graphical description of the classfication result. The system is made based on the machine learning model , EfficientNet, that is trained to learn the patterns of normal lungs states, pnuumonia, tuberculosis.

5.2 Functional and Non-Functional Requirements

System functional requirements are what the software is required to accomplish in order to achieve the desired functionality of the system unsupported file format or damaged images, and give a good feedback to the user. This guarantees that communication is also comfortable, and does not lead to confusion or abuse. Although the functional requirements are what the system does, the non-functional requirements are what determine the effectiveness with which the system handles those tasks.

Reliability helps to maintain the reliability of the system under various conditions. The model should yield consistent forecasts when fed with comparable inputs and the software should be able to run smoothly without frequent errors and crashes. This uniformity is needed in the creation of a trust among users, especially in an area where accuracy and reliability are the most important.

Scalability deals with the capability of the system to deal with growing workloads. The system must be able to support the increasing number of users or data without performance degradation with increased users or data. This is especially applicable when used within a large healthcare network or telemedicine platform and there can be many users of the platform at the same time.

The issues of security and privacy cannot be ignored in the process of dealing with medical data. The system should also provide security on the processing of the uploaded images as well as on the sensitive information of patients. Although the implementation can be performed in a controlled setting at present, the implementation in the future needs to comply with healthcare data protection standards to ensure that the trust of the users is not endangered.

Maintainability is the ease in which the system can be maintained or enhanced as time goes by. Since AI technologies evolve rather quickly, the software should be created in a modular form, so that the components of it (the model or interface) can be altered without affecting the whole system. This flexibility guarantees that the system will be up to date and able to absorb new developments in the future.

Table 5.1: Summary of Functional and Non-Functional Requirements

Requirement Type	**Description**	**System Implementation**
Functional	Image upload and preprocessing	Accepts X-ray images and prepares them for analysis
Functional	Disease prediction	Classifies images into Normal, Pneumonia, TB
Functional	Heatmap generation	Uses Grad-CAM for visual explanation
Functional	Result display	Shows prediction with confidence score
Non-Functional	Performance	Fast prediction with low latency
Non-Functional	Usability	Simple and intuitive interface
Non-Functional	Reliability	Consistent and stable outputs
Non-Functional	Security	Protects medical data integrity

Overall, the software requirement specification exposes a system that is keenly sensitive with regard to technological capacity and human requirement. It is not just a gathering of features but a unifying structure geared to convert the multifaceted computation into available and comprehensible results. The system strives to be a dependable part of the diagnostic process, silent yet efficient, accurate and consistently reliable, transparent and insightful, basing its functionality on actual clinical needs and molding its operation through intelligent design.

CHAPTER 6: DESIGN

6.1 System Architecture Overview

The proposed system architecture is designed with a purposeful compromise between the depth of computations and the clarity of operations, which are the two demands of the system in medical settings: accuracy and usability. The basic design of the system is a layered one with every part executing a specific task but is still linked in a single pipeline.

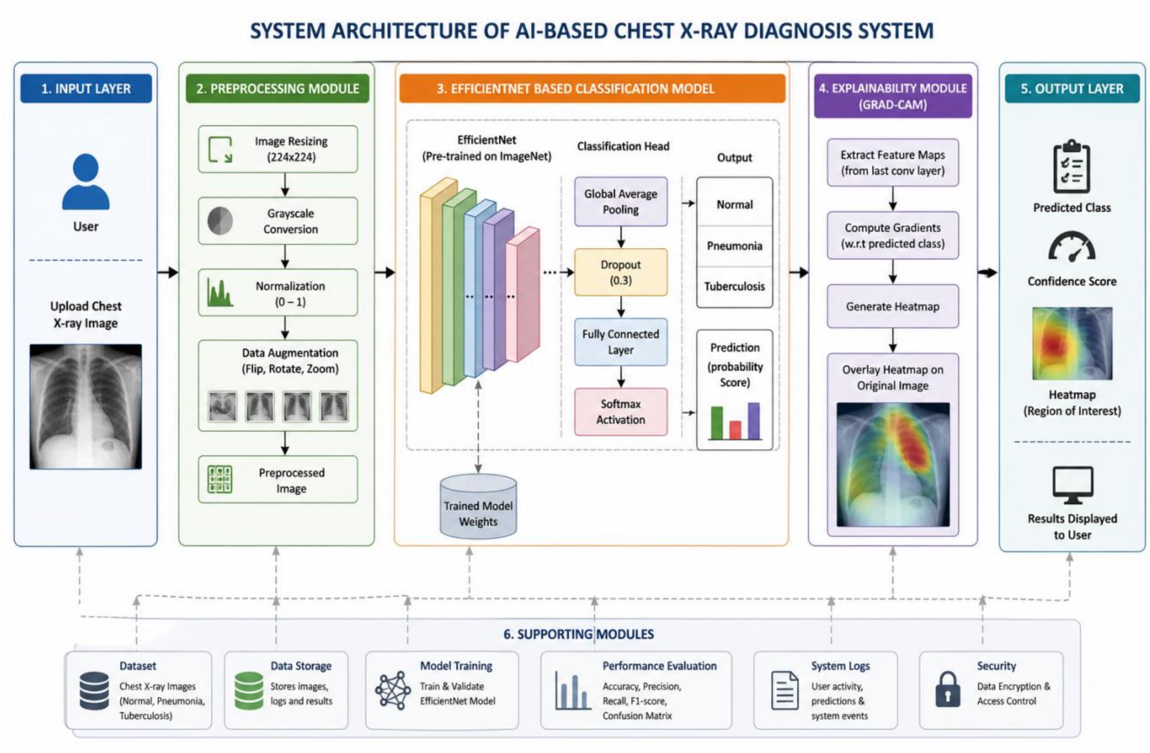


Figure 6.1: System Architecture of AI-Based Chest X-ray Diagnosis System

6.2 Data Flow Diagram

The flow of data within the system is systematic and chronological and therefore every transformation is meaningful and can be tracked.

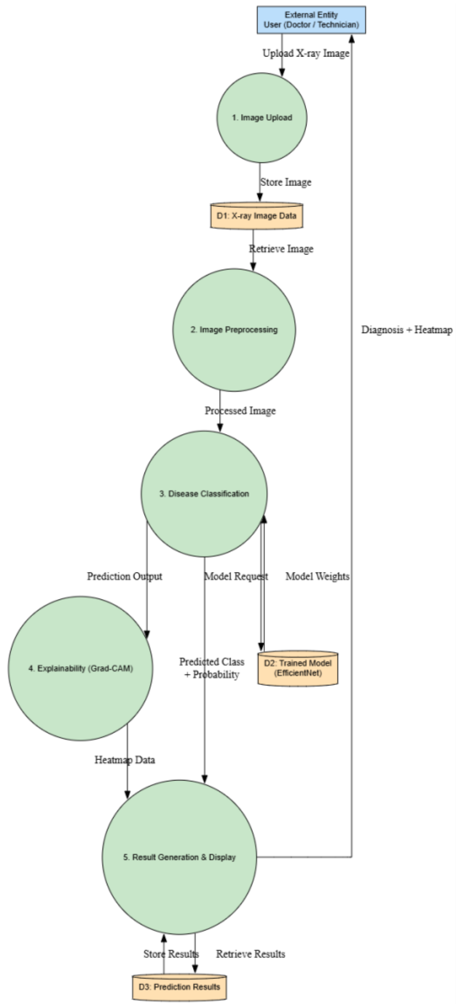


Figure 6.2: Data Flow Diagram (DFD Level 1) of the Proposed System.

6.3 Model Workflow.

The model workflow summarizes the logic of operation of the system, which involves how the data is processed between the input and output based on a sequence of calculations.

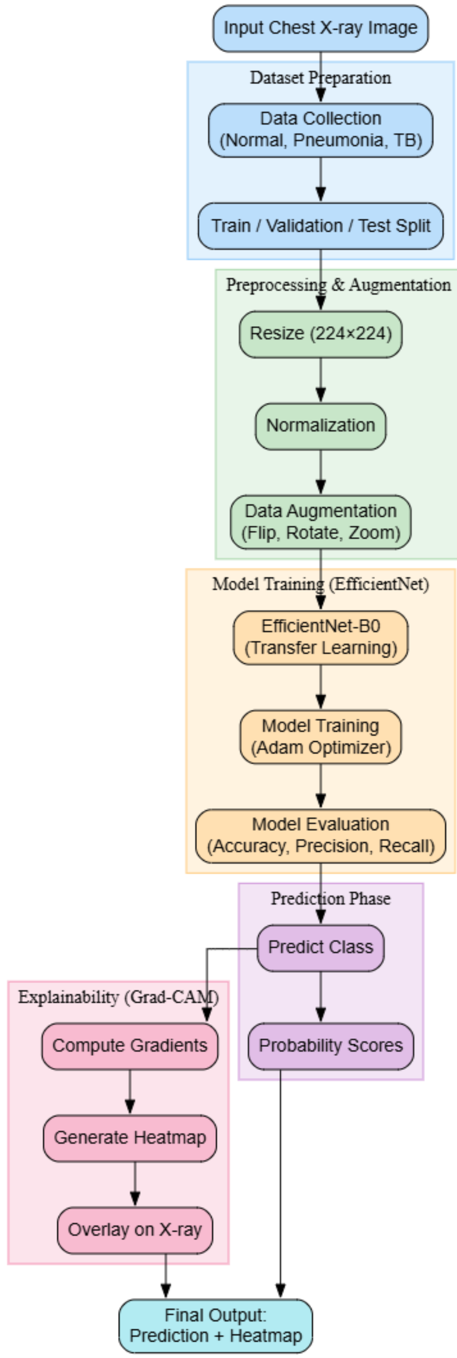


Figure 6.3: Chest X-ray Classification Model Workflow Diagram.

6.4 Pseudocode for Prediction

The logic of the system can be captured in the form of a structured sequence of operations that represent the prediction process. This pseudocode gives a clear explanation of the transformation of input data to output results.

```
Algorithm: Chest X-ray Disease Prediction

Input: X-ray Image I

Output: Predicted Class C, Confidence Score P, Heatmap H

Step 1: Load trained EfficientNet model M

Step 2: Preprocess input image I

    - Resize image to 224 × 224

    - Normalize pixel values

    - Convert to tensor format

Step 3: Pass preprocessed image into model M

    P = M.predict(I)

Step 4: Determine predicted class C

    C = argmax(P)

Step 5: Generate Grad-CAM heatmap

    - Compute gradients of class C with respect to final convolutional layer

    - Weight feature maps using gradients

    - Apply ReLU to obtain heatmap H

Step 6: Overlay heatmap H on original image

Step 7: Display predicted class C, probability P, and heatmap H
```

This structured logic ensures that each stage of prediction is both transparent and reproducible, reinforcing the reliability of the system.

Table 6.1: Module Description of the Proposed System

Module	**Description**	**Function**
Input Module	Handles image upload from user	Accepts chest X-ray images
Preprocessing Module	Standardizes image size and format	Resizing, normalization
Model Module	EfficientNet-based classifier	Feature extraction and prediction
Explainability Module	Grad-CAM integration	Generates heatmaps
Output Module	Displays results to user	Shows prediction and visualization

CHAPTER 7: TESTING

Table 7.1: Performance Metrics of the Chest X-ray Classification Model

Class	Precision	Recall	F1-Score	Support
Normal	0.91	0.75	0.82	234
Pneumonia	0.86	0.97	0.91	390
Tuberculosis	1.00	0.98	0.99	517
Overall Accuracy	—	—	0.93	1141

The values shown in the table depict the balanced performance of the model as the weighted F1-score is more than ninety percent. The results of this nature indicate that the system can be applied to be consistent across classes as well as mitigate the issues related to class imbalance and feature similarity.

When considering the testing phase, it is possible to see the numerical success besides a more profound correspondence between the model behavior and clinical expectations. The system is shown to be able to recognize patterns of disease with accuracy, to justify its reasoning in a way that is understandable visually and that it will work consistently in a practical system. However, testing is a way to also see the fine lines of performance, and to be reminded that no model is perfect, and that constant improvement is a key element of performance. The results, in turn, do not simply confirm the system, they shed light on its strong points, admit its weak ones, and spur its further development in the constantly challenging environment of medical diagnostics.

7.1 Key Implementation Details

```
IMG_SIZE = (224, 224)

BATCH_SIZE = 32

SEED = 123

train_ds = tf.keras.preprocessing.image_dataset_from_directory(

    train_dir, label_mode="int", image_size=IMG_SIZE, batch_size=BATCH_SIZE, seed=SEED

)

val_ds = tf.keras.preprocessing.image_dataset_from_directory(

    val_dir, label_mode="int", image_size=IMG_SIZE, batch_size=BATCH_SIZE, seed=SEED

)

test_ds = tf.keras.preprocessing.image_dataset_from_directory(

    test_dir, label_mode="int", image_size=IMG_SIZE, batch_size=BATCH_SIZE, seed=SEED, shuffle=False

)

class_names = train_ds.class_names

num_classes = len(class_names)

print("Classes:", class_names)


AUTOTUNE = tf.data.AUTOTUNE

train_ds = train_ds.prefetch(AUTOTUNE)

val_ds = val_ds.prefetch(AUTOTUNE)

test_ds = test_ds.prefetch(AUTOTUNE)

base_path = "/content/drive/MyDrive/Chest_XRAY_DATASET/"

splits = ["train", "test", "val"]

classes = ["NORMAL", "PNEUMONIA", "TUBERCULOSIS"]
```

```
def show_samples_all_sets(base_path, splits, classes, samples_per_class=3):
    for split in splits:
        print(f"\n===================== {split.upper()} SAMPLES =====================\n")
        plt.figure(figsize=(12, 10))
        img_index = 1
        for cls in classes:
            class_path = os.path.join(base_path, split, cls)
            imgs = os.listdir(class_path)

            # If folder has fewer images, adjust sampling
            selected = random.sample(imgs, min(samples_per_class, len(imgs)))
            for img_name in selected:
                img_path = os.path.join(class_path, img_name)
                img = cv2.imread(img_path)
                img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
                plt.subplot(len(classes), samples_per_class, img_index)
                plt.imshow(img)
                plt.title(f"{split.upper()} - {cls}")
                plt.axis("off")
                img_index += 1

        plt.tight_layout()
        plt.show()
```

```
# ✅ Run Function
show_samples_all_sets(base_path, splits, classes)


from tensorflow.keras.preprocessing.image import load_img, img_to_array
from tensorflow.keras.applications.efficientnet import preprocess_input


def gradcam_rebuild_head_no_conf(img_path, full_model, class_names, target_size=(224,224)):
    print("\n===== GRAD-CAM (No Confidence) START =====")


    # ---- Load & preprocess ----
    pil = load_img(img_path, target_size=target_size)
    arr = img_to_array(pil).astype("float32")
    original_uint8 = arr.astype("uint8")
    x = preprocess_input(arr.copy())
    x = np.expand_dims(x, axis=0)
    x_tf = tf.convert_to_tensor(x)


    # ---- Find EfficientNet backbone ----
    try:
        backbone = full_model.get_layer("efficientnetb0")
        print("✅ Found backbone layer:", backbone.name)
    except:
        print("❌ EfficientNet not found.")
        return
```

```
# ---- Last Conv ----
last_conv_name = None
for layer in reversed(backbone.layers):
    if isinstance(layer, tf.keras.layers.Conv2D):
        last_conv_name = layer.name
        break

if last_conv_name is None:
    print("❌ No Conv2D found inside backbone.")
    return

print("✅ Last Conv Layer:", last_conv_name)
conv_layer = backbone.get_layer(last_conv_name)

# ---- Rebuild classifier head ----
layer_names = [l.name for l in full_model.layers]
backbone_index = layer_names.index(backbone.name)
head_layers = full_model.layers[backbone_index+1:]

x_head = backbone.output
for L in head_layers:
    x_head = L(x_head)

head_output = x_head
```

```
# ---- Build grad model ----
grad_model = Model(
    inputs=backbone.input,
    outputs=[conv_layer.output, head_output]
)

# ---- Compute Gradients ----
with tf.GradientTape() as tape:
    conv_maps, preds = grad_model(x_tf, training=False)

    # ✅ Tell tape to watch convolutional outputs
    tape.watch(conv_maps)

    if preds.shape[-1] > 1:
        pred_index = tf.argmax(preds[0])
        score = preds[:, pred_index]
    else:
        pred_index = tf.cast(tf.math.round(tf.sigmoid(preds[0][0])), tf.int32)
        score = preds[:, 0]

grads = tape.gradient(score, conv_maps)

if grads is None:
    print("❌ Gradient returned None — stopping.")
    return
```

```
grads = grads[0].numpy()
conv_out = conv_maps[0].numpy()

# ---- CAM Construction ----
weights = np.mean(grads, axis=(0,1))
cam = np.dot(conv_out, weights)
cam = np.maximum(cam, 0)
cam /= (np.max(cam) + 1e-9)

cam_resized = cv2.resize(cam, (target_size[1], target_size[0]))
heatmap = cv2.applyColorMap(np.uint8(cam_resized * 255), cv2.COLORMAP_JET)
heatmap = cv2.cvtColor(heatmap, cv2.COLOR_BGR2RGB)
overlay = cv2.addWeighted(original_uint8, 0.6, heatmap, 0.4, 0)

# ✅ Only class name — NO confidence
pred_label = class_names[int(pred_index)]

# ---- Show Output ----
plt.figure(figsize=(11,5))
plt.subplot(1,2,1)
plt.title("Original Image")
plt.imshow(original_uint8.astype("uint8"))
plt.axis("off")

plt.subplot(1,2,2)
plt.title(f"Grad-CAM → Predicted: {pred_label}")
plt.imshow(overlay.astype("uint8"))
```

```
    plt.axis("off")
    plt.show()

    print("✅ Predicted:", pred_label)

    return

# ---- Run Test ----
img_path = "/content/drive/MyDrive/Chest_XRAY_DATASET/test/NORMAL/IM-0001-0001.jpeg"
gradcam_rebuild_head_no_conf(img_path, model, class_names)
```

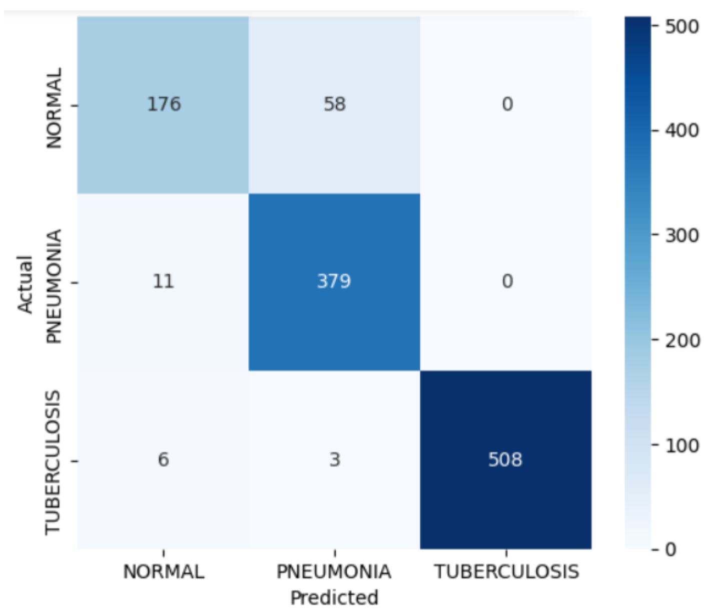


Figure 7.1: the confusion matrix of the proposed deep learning model, showing its classification performance across three classes—Normal, Pneumonia, and Tuberculosis—illustrating how accurately the model distinguishes between these categories.

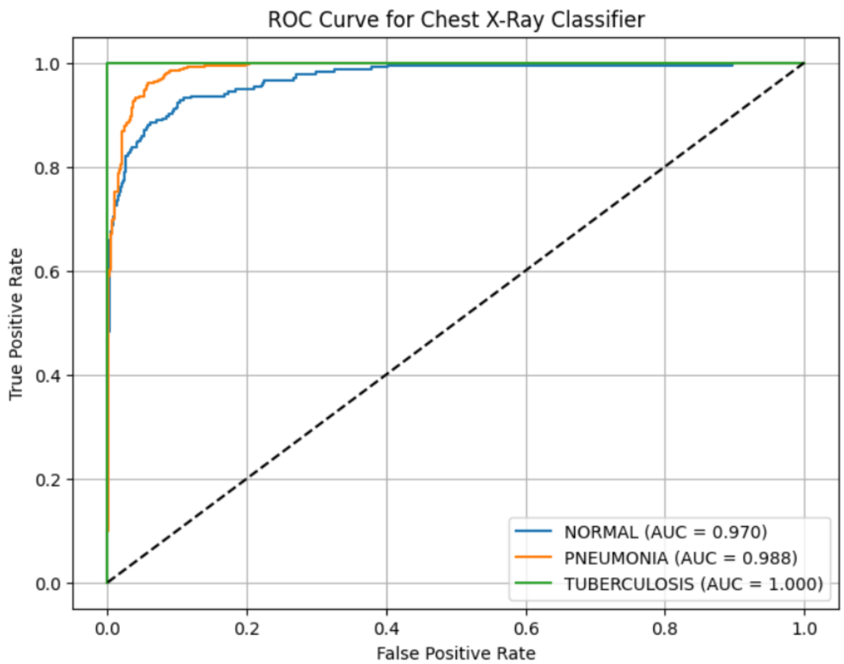


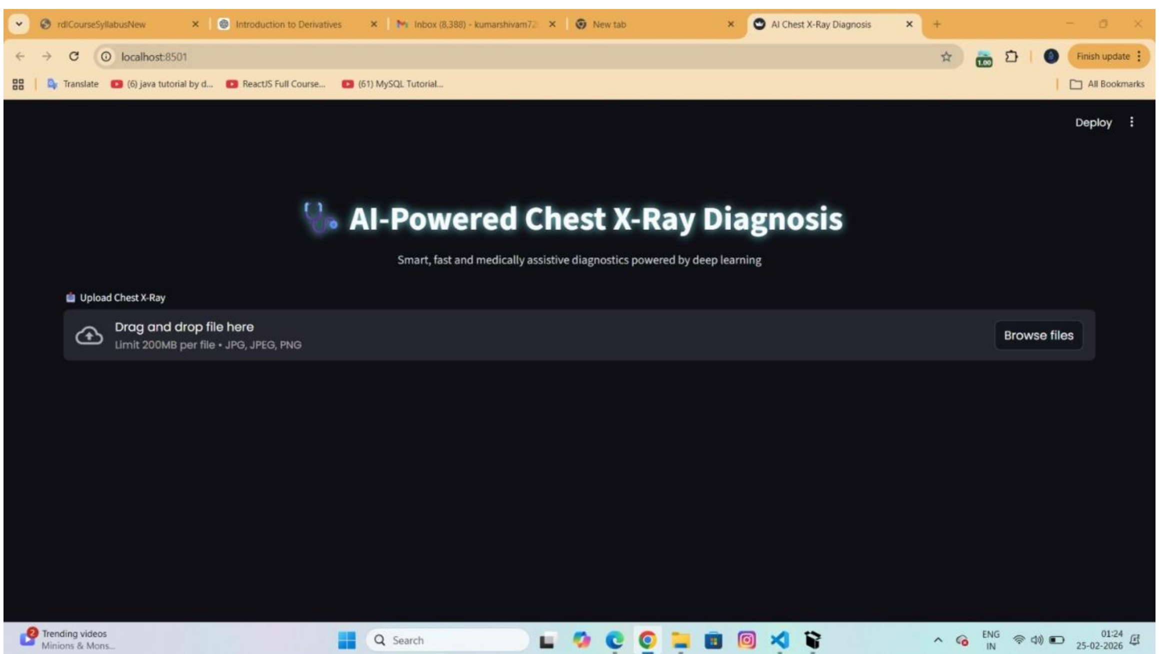
Figure 7.2: ROC curves of the proposed chest X-ray classifier with high AUC values for Normal (0.970), Pneumonia (0.988), and Tuberculosis (1.000).

CHAPTER 8: IMPLEMENTATION

8.1 System Implementation Overview

The introduction of the suggested system can be considered the shift towards the conceptual design to a functioning diagnostic tool that can help in real-life medical analysis. It is here that abstract models and theoretical processes are converted into a practical application where each part such as data processing, model inference and user interaction all works in concert. The system is created with a mixture of Python-based frameworks, which were selected based on their technical capacity, as well as their reliability and popularity in the artificial intelligence domain.

Essentially, the implementation is in a modular format. The data processing and model inference are done by the backend, whereas the interaction with the user takes place through a user-friendly interface in the frontend. This division makes it so that the user will never see the computational complexity of the system and the results provided by the system in a clear and understandable form. The design is focused on being easy to use without the need to lose functionality and as a result, even individuals who lack technical know how can easily use the system.



8.2 Technologies Used

Its implementation is based on a well-chosen set of technologies which meets the needs of deep learning and web-based implementation. Python is the main programming language, which is flexible and has a wide array of libraries. During development and training of models, it will use TensorFlow and Keras, which offer a powerful platform to develop and optimize neural networks. Such frameworks allow processing large amounts of data effectively, facilitate the use of GPUs, and provide pre-trained models like EfficientNet, which greatly helps save time on development.

Image processing operations are done through libraries like OpenCV and NumPy which help to do operations like resizing, normalization, array manipulation etc.

8.3 Model Training Process

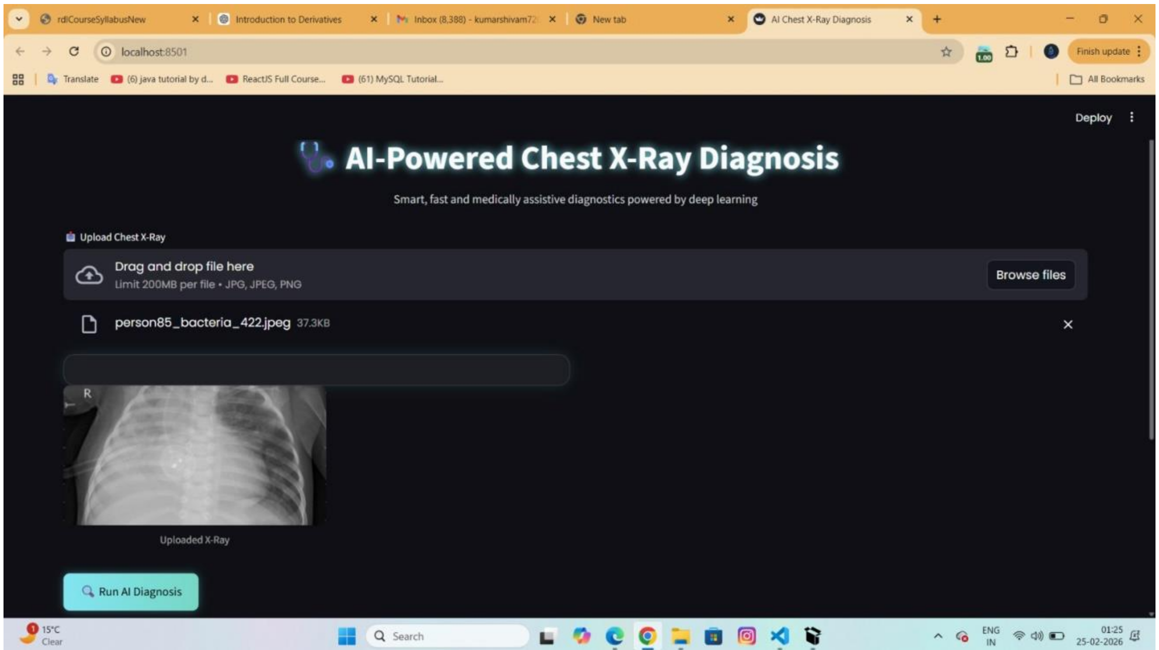
The basis of the predictive capability of the system is created through the training process. It starts with the preparation of datasets, in which chest X-ray images have been gathered and divided into three groups: normal, pneumonia, and tuberculosis. The data is then separated into training, validation and testing sets to have an unbiased assessment.

8.4 Deployment

Deployment stage converts the trained model into an application. Streamlit is a web-based system that is used to deploy the system and is accessible via a browser. Upon opening the application, the user is greeted with a clean interface, where he/she is shown the opportunity to upload a chest X-ray picture. The image uploaded is processed in real time and the results are shown instantly after undergoing the trained model.

The backend combines the trained TensorFlow model, which is the prediction and Grad-CAM creation. Once an image has been posted by a user, the system will preprocess it and feed it into the model. The confidence score and predicted class are calculated and a heatmap created with Grad-CAM. This heatmap is superimposed to the original picture, showing the areas that affected the prediction.

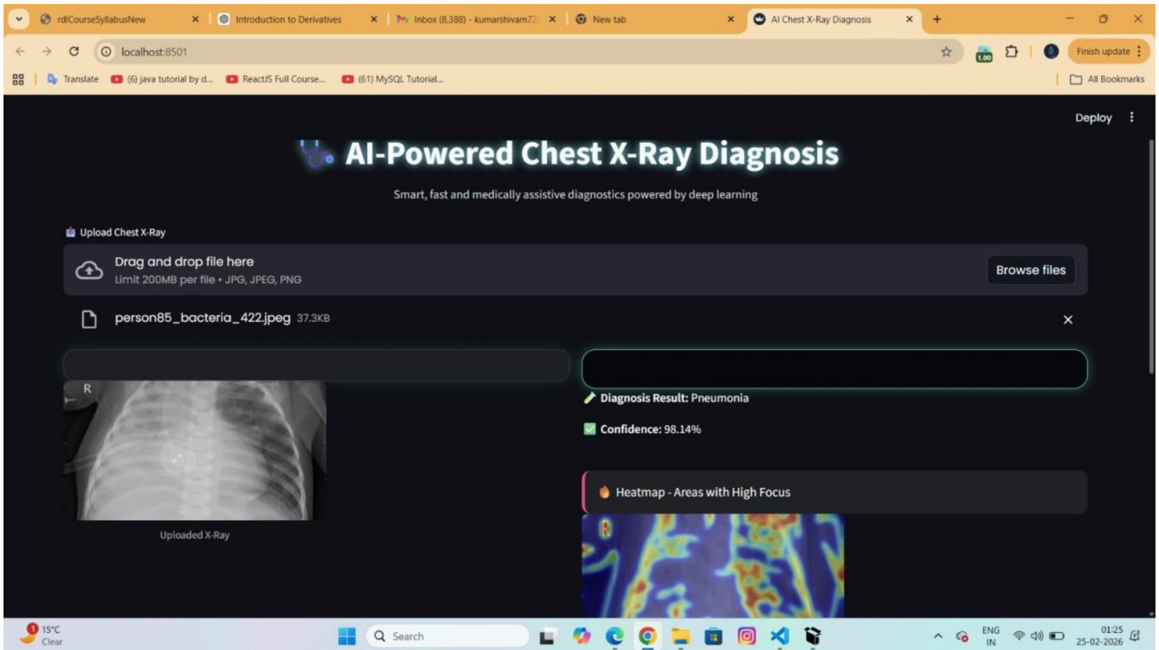
The model is transformed to TensorFlow Lite to enhance performance and portability. It minimizes the size of the models and the inference time and can be deployed to low-resource devices. It is especially useful in applications to rural healthcare or telemedicine, where a state-of-the-art computational infrastructure might not exist.



The implemented system has a low latency where forecasts are made in seconds. This quickness is paramount in the clinical environment, and prompt decision making can be of great importance to the patient. The interface will be user-friendly, with clear instructions and few steps of interaction, so that the users will be able to use the system without any problems.

8.5 User Interface Integration

The user interface is the interface that links the system and the users. It is created to be simple to the eye, but complete in functionality and takes users through the procedure of uploading a picture and seeing the results. The interface will have a drag and drop upload option, which will enable users to choose and post X-ray images easily. After the image is uploaded, the system automatically processes the image and there is no need of manual intervention.



The results are presented in a well-organized format, such as the predicted class, confidence score, and the Grad-CAM heatmap. Visual elements make the process of understanding better, as a user is able to interpret results effectively and quickly. Another aspect that the interface guarantees is that the feedback is instantaneous, which supports the responsiveness and reliability of the system.

CHAPTER 9: PROJECT LEGACY

No project, when finally completed, is left behind just a working system; it bears with it a history of decisions, lessons, refinements and even silent realizations which define future work. The creation of the AI-based system of diagnosis of chest X-rays is not an exception. What started as a technical exploration of deep learning slowly turned into a more in-depth exploration of the reality of healthcare, where accuracy is not just good but a requirement, and each forecast can have a consequence. The heritage of this project can thus be seen to go beyond the end result to the perception it has created and the path it proposes to take.

The system is at stage of a working prototype, and it is able to carry out multi-classification of X-ray images of the chest with high degree of accuracy. It involves the use of EfficientNet to extract and classify the features with the usage of preprocessing pipelines that maintain consistency in the input data. Grad-CAM has enabled the transformation of the system into an interpretable assistant and provide visual explanations, consistent with clinical reasoning. This transparency and performance is a deliberate attempt to leave behind the traditional black-box nature of deep learning in an understanding that in medicine, trust is as important as precision. The interface is also facilitated by the use of a Streamlit-based interface, which further enhances its usability, allowing immediate interaction and without the need to have specialized technical knowledge.

However, the existing system also shows the limits, on which it performs.

It has demonstrated that the marriage of artificial intelligence and medical imaging is challenging and potentially a successful business, but needs to be carefully balanced between technological and human capacity. It has brought out the need to be clear, reliable and ethical in designing diagnostic systems. Above all, this has confirmed that the real progress in this area is not dictated by a single breakthrough but a gradual, well-considered evolution- where every step, regardless of how small, goes towards a greater picture of affordable and reliable healthcare.

CHAPTER 10: USER MANUAL

The user manual is an effective guide to the process of using the AI-based chest X-ray diagnosis system, where a set of technical functions are transformed into easily understandable and easy-to-use steps. The system is also designed in such a way that it is simple and those with little technical background can use the system effectively. These operations, though they are going on in the background are critical to the accuracy of the system. The opportunities to give visual explanations with the help of Grad-CAM are one of the peculiarities of the system.

It is also worth noting that the system is rather designed as a decision-support system and not to substitute professional medical judgement. Although it provides precise forecasts and useful information, it is a matter of qualified medical practitioners to make the ultimate diagnosis and conclusion. The purpose of the system is to help, to point out trends, and to offer one more viewpoint that can potentially benefit clinical decisions.

To conclude, the user manual is based on the design philosophy that is focused on the accessibility, clarity, and reliability. Every step, including image upload, and interpretation of the results, has been designed in a manner that makes the system user-friendly, yet capable of providing useful diagnostic assistance. With cutting-edge computing power and a simple user interface, the system fills the gap between sophisticated technology and the real-world clinical setting and enables users to interact with artificial intelligence in a way that is both useful and informative.

CHAPTER 11: CONCLUSION & FUTURE WORK

The article in this project shows how a well-developed deep learning system can be a convenient and helpful tool in decoding the chest X-ray image.

It would be possible to optimise the system to work on the mobile and edge devices to be used in the remote or resource-limited settings where access to expert radiological services is frequently low. Automated report generation and integration with hospital information systems might also be added to the system, further streamlining clinical workflows, turning the system into a part of a broader healthcare ecosystem, rather than just a diagnostic tool.

To sum up, the given project is a significant move towards the implementation of artificial intelligence in the medical imaging field. It emphasizes the possibility as well as the accountability that goes along with such progress. The system provides a platform on which future innovations can be built as it offers accuracy coupled with interpretability and accessibility, and helps to create a vision of healthcare that is more responsive, informed, and accessible to many.

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